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GENERALITY OF SPOKEN DIALOGUE SYSTEM USING SEGA-IL FOR DIFFERENT LANGUAGES

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ABSTRACT
This paper presents a Spoken Dialogue Method Using Inductive Learning Method Based on Genetic Algorithm with Sexual Selection (SeGA-ILSD) along with analysis of its generality for different languages: Japanese and English. This method achieves high generality using a learning function. Therefore, this method can show equivalent performance for different languages. To confirm it, we developed an experimental system of this method and carried out an experiment of performance evaluation of this method. These results show that this method has equivalent performance for two languages: Japanese and English. Results verify that this proposed method has high generality for these languages.

KEY WORDS
Genetic algorithm, Inductive Learning, spoken dialogue processing, sexual selection, and generality

1. Introduction

Early in the history of computer development, research into dialogue processing began. For example, ELIZA [1] is a typical and very famous dialogue system that uses a keyword and obtains information from humans through interviews. Importantly, ELIZA can accept any utterance. For that reason, some users believe that ELIZA's responses are from a human. However, it is too difficult for ELIZA to generate an adaptive reply to a user's utterance because it uses only static rules. For that reason, ELIZA can not satisfy most users. In contrast, SHRDLU [2] can understand natural language, but it can talk with users only about a small block world. Because SHRDLU has all knowledge about the small block world, it can answer any question about it, but only about it. Therefore, SHRDLU also cannot satisfy users. Colby et al. [3] developed a dialogue system for treatment of depression. This system uses a thesaurus, a semantic database, coincidence, and other processes. In addition, it generates a reply using a production rule. It is effective for treatment of depression, but this system also uses a manually calculated static rule. Therefore, we must take huge labor and costs in case the system applies to a different purpose.

Many researchers have developed spoken dialogue systems for chat [4]. However, such systems require much time and effort initially for formulating many reply sentence generation rules. Therefore, we have already developed a Spoken Dialogue Method Using Inductive Learning with Genetic Algorithm [5]. We call it GA-ILSD. The GA-ILSD is a chat system and can learn language from dialogue examples. This method has a learning function. However, the performance of this method is insufficiently high for any language because the generality of this method is insufficient.

Therefore we created SeGA-ILSD by applying a genetic algorithm with sexual selection to GA-ILSD. This method generates reply generation rules automatically from pairs of an utterance and its associated reply. This method can generate these rules even though an initial dictionary is empty. This means that a system based on this method has no language information, such as vocabulary and grammar as the initial state. Only this system has a learning function. For that reason, it has high generality for various languages.

In this paper, we describe an algorithm of SeGA-ILSD and a performance evaluation experiment to confirm the generality of this method for Japanese and English. Results of this experiment show that SeGA-ILSD has the same performance for these languages. Therefore, we confirmed that SeGa-ILSD has high generality for these languages.

2. Outline of GA-ILSD

Systems based on GA-ILSD acquire rules automatically from pairs of an utterance and its associated reply. GA-ILSD includes two processes: ELIZA and GA-IL.

Systems based on ELIZA use a keyword. In this method, the system tries to find a sentence that includes the same word as a keyword. If it finds its word, it presents a corresponding sentence as a reply. However, ELIZA's reply is always ambiguous. Therefore, it can only chat.

In contrast, GA-IL can learn a reply generation rule. It includes an actual reply because GA-IL can achieve acquisition from actual pairs of utterances and replies.

Initially, GA-ILSD has no linguistic knowledge, i.e., vocabulary and syntactic information. It has only a generation process using ELIZA and GA-IL. Before learning, GA-ILSD has no reply generation rules produced by GA-IL. Therefore, it replies only using ELIZA rules.

The GA-ILSD system acquires reply generation rules from the pairs of a system's utterance and its user's reply. ELIZA and GA-IL generate the system's utterance.

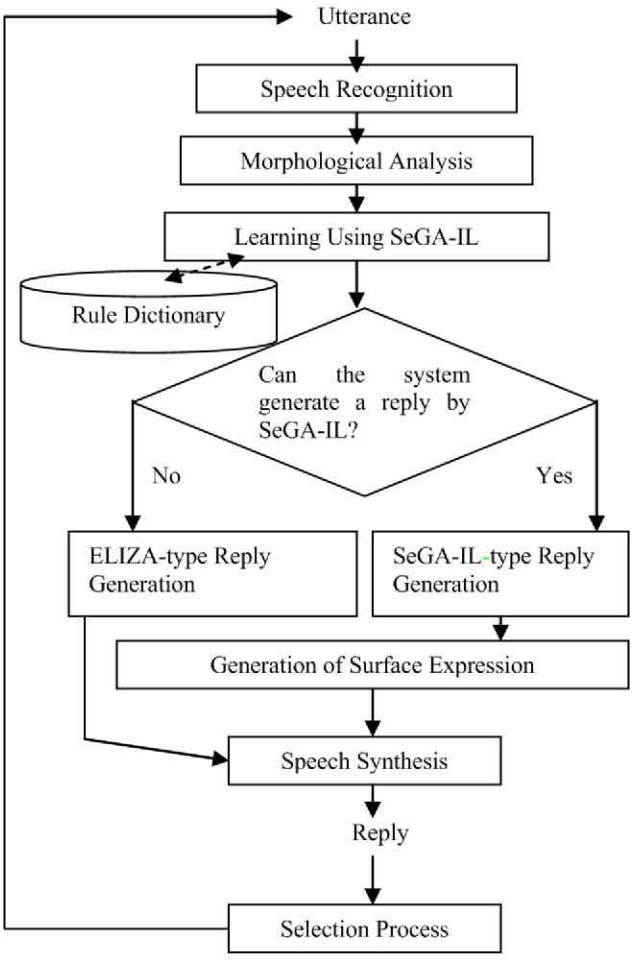


Fig. 1 Process of SeGA-ILSD

In GA-ILSD, the correct rate is used to judge fitness. Therefore, the system cannot calculate the fitness of a rule unless the rule is used. This rule is never deleted if the system does not use it. Consequently, too many bad rules remain. For that reason, the rate of correct responses remains low. We have proposed SeGA-ILSD to resolve this problem.

3. Genetic Algorithm with Sexual Selection

An evaluation with sexual selection differs from natural selection. Natural selection is based on survival in an environment. However, this theory cannot explain a peacock's feathers or a lion's mane. Darwin proposed sexual selection to explain differences by sex [6]. We call it SeGA. Natural selection evolves characteristic form and quality to adapt to an environment. On the other hand, sexual selection compels them to select a mate.

A female exerts more labor to obtain children than a male because she must carry out egg-laying, foster gestation, and perform child-rearing. Therefore she must select a superior male to provide her superior children: many females select a superior male [7].

4. Algorithm of SeGA-ILSD

In SeGA-ILSD, each reply generation rule has a sexual distinction. A female rule prefers a superior male rule. In this case, the preference of the female rule is "to generate rules that include many common parts."

Therefore a system based on GA-ILSD evolves into increased learning of rules. The system can generate a superior rule efficiently because it can learn rules from examples even if they include only a short common part.

4.1 Outline

Figure 1 depicts the SeGA-ILSD process. A user utters something. First, the system recognizes the utterance. This system uses Japanese Recognizer ver. 6.1 (Microsoft Corp.) and English Recognizer ver. 5.1 (Microsoft Corp.), respectively, as speech recognition tools for Japanese and English. This system also uses LH Kenji Version 5.1(Microsoft Corp.) and Mike Version 5.1 (Microsoft Corp.), respectively, as speech synthesis tools for Japanese and English.

Next, the system carries out morphological analyses for the recognition result. The system uses JUMAN Version 5.0 [8] and Apple Pie Parser Version 5.9 [9], respectively, as Japanese and English morphological analysis tools.

The system extracts a reply generation rule from among pairs of a user's utterance and its system's reply, which are registered in a rule dictionary. The rule dictionary has pairs of a user's utterance and the system's reply and reply generation rules that the system extracts until that time.

The system judges whether it can generate a reply using a SeGA-IL-type rule or not. It uses ELIZA-type reply generation if it cannot generate a reply using the SeGA-IL-type rule. Before learning, the system has insufficient SeGA-IL-type rules. Therefore, it uses ELIZA-type reply

generation. However, in later stages of learning, it has sufficient SeGA-IL-type rules generated by SeGA-IL. Therefore, it generates a reply using SeGA-IL-type reply generation.

Subsequently, the system generates surface expressions only for Japanese because it deletes a part of speech and so on at the learning using SeGA-IL.

The system uses speech synthesis tools to generate a voice for the reply. Finally, the system carries out the selection process. In this step, it deletes reply generation rules using a fitness value. In this system, we use the correct response rate as the fitness value.

4.2 SeGA-IL-type Rule Generation

In this process, the system extracts an independent word for Japanese because the independent word includes meanings in Japanese. Moreover, numerous errors exist in parts of speech and so on in speech recognition results because their length is too short.

4.2.1 Generation of Examples Using SeGA

In SeGA, a rule corresponds to a chromosome, and a word of the rule corresponds to a gene. Rules are of two kinds. One has no variable, another includes a variable. The variable means that any word can replace this symbol. Inductive Learning produces the variable. A rule with a variable has higher generality than a rule without a variable. For this method, we define a male rule as a rule with a variable; a female rule lacks a variable.

The male rule must ornament itself to attract a female rule. Therefore the male rule must include a variable because the variable can be replaced by a word included in an utterance. In Fig. 2, (1), (2), (4), (5), (6), (8) and (10) are female rules, and (3), (7) and (9) are male rules. Female rules include original pairs of an utterance and its reply.

In this method, the benefit of a female rule is that a male rule includes more common parts with its female rule. In this definition, the direction of evolution of a rule tends toward increased rules.

In the next step, the female rule selects a preferred male rule from among the ornamented male rules. Subsequently, the system undertakes uniform crossover, by which the system randomly generates a crossover mask. To avoid destroying the sentence structure, the system uses this crossover mask only for cases in which both parents have a common part of speech.

4.2.2 Ornateness

A male rule replaces its variable with a female rule that does not include a variable. According to this operation, the male rule tries to find a female rule that has more common words, thereby obtaining many "children".

An example of the ornateness is as follows:
(1) A male rule: Could you @1 @2 @3 question please
(2) After ornateness: Could you answer my previous question please

In this example, "@" denotes a variable. The system replaces "@1 @2 @3" with "answer my previous". The reason is that some female rules include "answer my previous."

4.2.3 Inductive Learning

After the system increases examples using SeGA, it generates a rule using Inductive Learning [10], through which the system compares two rules, and then extracts a common part and a different part recursively. The system registers these rules in a rule dictionary.

[I] Pairs of an utterance and its reply
(1) Do you play tennis: Yes I like tennis
(2) Do you play soccer: Yes I like soccer.
[II] Rules generated from [I]
(3) Do you play @1: Yes I like @1 (4) tennis: soccer
[III] Other pairs of an utterance and its reply
(5) Do you eat apples: Yes I eat apples
(6) Do you eat it: Yes I eat it
[IV] Rules generated from [III]
(7) Do you eat @1: Yes I eat @1 (8) apples: it
[V] Rules generated from (3) and (7)
(9) Do you @2 @1: Yes I @2 @1 (10) play: eat

Fig. 2 Rule Generation Using Inductive Learning

Figure 2 shows an example of rule extraction using Inductive Learning. In Fig. 2, (1) and (2) are pairs of one utterance and one reply. First, the system generates rules (3) and (4) from them. Rule (3) is generated by replacing the different part with "@1". Rule (4) is generated by pairs of different parts between (1) and (2). Also, (5) and (6) are other pairs of an utterance and its reply. Similarly, rules (7) and (8) are generated from (5) and (6). Moreover, rules (9) and (10) are generated from rules (3) and (7) in the same way, recursively.

4.3 Generation of SeGA-IL-type Reply

The system selects the most suitable rules for a recognition result from among SeGA-IL-type rules. Concretely, the conditions of selection of the most suitable rules are as follows:

(1) The rule requires that the rate of identical words between independent words in an utterance and that in a left side of a rule be greater than 65%.
(2) The rule must have the highest rate of fitness value described in 4.7.

4.4 Generation of an ELIZA-type Reply

The system is designed to generate a reply using an ELIZA-type rule if the system cannot find a SeGA-IL-type rule that satisfies the above-mentioned conditions.

In this process, the ELIZA-type rule is generated in advance by a human. They are 400 for Japanese, and 461 for English, with 167 and 48 keywords, respectively.

4.5 Generation of Surface Expression

The reply generated by SeGA-IL-type rules has only an independent word in Japanese. Therefore, the system must translate to a natural expression. In this process, it adds independent word strings to an adjunct.

The system uses surface expression generation rules to generate natural expressions. The surface expression generation rule is generated from pairs of an independent word string and its natural expression in the same manner as for the reply generation process using SeGA-IL.

4.6 Selection Process

A user evaluates a reply by this evaluation standard:

(1) Correct reply: Meaning is correct, and expression is natural.
(2) Semi-correct reply: Meaning is correct, but expression is not natural.
(3) Erroneous reply: Meaning is not correct.

The rules used for generating this reply are updated using the decision. The system increases the correct reply frequency for the rules that are used to generate the correct reply and the semi-correct reply. In addition, it increases the erroneous reply frequency for the rules that are used to generate the erroneous reply. Moreover, the system increases the unused frequency for the rules that are not used. A rule is deleted from the rule dictionary if a fitness value for a rule calculated by an eq. (1) in 4.7 is less than 75.

4.7 Conditions of GA

A definition of the fitness value is shown as eq. (1).

$$FitnessValue = \frac{A}{A+B} \times \alpha \times 100 \qquad (1)$$

A: Correct reply frequency
B: Erroneous reply frequency

$$\alpha = \begin{cases} 1 : C \\ 1 - 0.08 \times UF : D \end{cases}$$

In that equation, C indicates a value for a dialogue example, UF means unused frequency, and D denotes a value for a rule generated using SeGA and Inductive Learning.

In eq. (1), "A" does not include the semi-correct replies. Thereby, it yields a more correct rule after a selection process. The maximum number of recursions is 10. The system learns from the last 30 extracted rules to reduce the CPU time. The mutation rate is 2%.

5. Experiments for Performance Evaluation

5.1 Conditions

The size of dialogues is 250 turns. As an initial state, the system has no SeGA-IL-type rules and an ELIZA-type rule. Five Japanese subjects participated. They are Japanese science postgraduate students. We call them A, B, C, D and E. All are male. One English subject participated. He is a Polish science postgraduate student called F herein.

5.2 Results and Considerations

5.2.1 Speech Recognition Rate and Reply Rate

Speech recognition rates were 86.1% for Japanese and 62.0% for English. Table 1 shows reply rates for Japanese. The reply rate shows the rate of reply generated using an ELIZA-type rule or SeGA-IL-type rules. We respectively call them ELIZA-type and SeGA-IL-type replies.

Among the results for Japanese, the reply rates of A, B, C, and E were almost equal. However, only the correct reply rate was different. The kinds of independent words of D were 1.5 times those of the others. Therefore, the speed of learning of D was slower than for the others.

The reply rates for English were 64.8% for the ELIZA-type replies, and 35.2% for the SeGA-IL-type replies. On the other hand, the average reply rate for Japanese was 73.9% for ELIZA-type replies, and 26.1% for SeGA-IL-type replies. These results show that the system performance is almost equal for both Japanese and English.

5.2.2 Precision

We next evaluate whether a reply is correct or not. Our standard is as follows:

(1) A correct reply generated by an ELIZA-type rule
(2) A correct reply generated by a SeGA-IL-type rule
(3) A semi-correct reply generated by an ELIZA-type rule
(4) A semi-correct reply generated by a SeGA-IL-type rule

(5) An erroneous reply generated by an ELIZA-type rule
(6) An erroneous reply generated by a SeGA-IL-type rule

Table 1 Reply Rate for Japanese

Subject	ELIZA-type reply [%]	SeGA-IL-type reply [%]	Number of kinds of independent words
A	73.2	26.4	348
B	70.8	29.2	328
C	71.7	28.3	343
D	82.0	18.0	499
E	71.6	28.4	347
Average	73.9	26.1	373

Table 2 Precision for Japanese

Subject	(1)	(2)	(3)	(4)	(5)	(6)
A	33.1	12.7	7.6	2.0	32.8	11.6
B	44.8	20.0	15.6	2.0	10.4	7.2
C	36.3	13.9	13.1	3.6	22.4	10.7
D	40.8	11.2	20.0	2.4	21.2	4.4
E	36.0	15.2	11.2	2.4	24.8	10.8
Average	38.2	14.6	13.5	2.5	22.3	8.9

[%]

Table 3 Precision for English

Subject	(1)	(2)	(3)	(4)	(5)	(6)
F	8.0	4.0	53.2	16.0	3.2	15.2

[%]

Tables 2 and 3 respectively show precisions for Japanese and English. We define that an effective reply rate is the total of (1), (2), (3) and (4). The effective reply rate is 68.8% for Japanese and 81.2% for English. Therefore, this method is effective for both languages.

The total of (2) and (4) shows an evaluation only for SeGA-IL. We define the effective reply rate for SeGA-IL as that sum, which was 17.1% for Japanese, and 20.0% for English. The SeGA-IL displays equivalent performance for both languages.

The effective reply rate of GA-ILSD is 59.3% for Japanese [2], meaning that the effective reply rate of SeGA-ILSD is 9.5 points higher than that of GA-ILSD. This result shows that SeGA is more effective for a spoken dialogue system for chat than GA.

5.2.3 Consideration of Dialogue Results

Figure 3 and Figure 4 respectively show examples of dialogues in Japanese and in English.

Figure 3 shows a Japanese example. At U1 and S1 of Fig. 3, a user utters "An apple's candy is delicious.", and the system replies "Maybe it is delicious." This reply is generated by an ELIZA-type rule. This conversation is very natural. Demonstrably, the ELIZA-type rule is excellent in Japanese.

U1: *Ringono Ame Oishiiyo*
(Apple candy is delicious.)
S1: *Oishiikamone* {ELIZA: correct}
(Maybe it is delicious.)
U2: *Mochirondesuyo* (Of course.)
S2: *Atsuito Kangaeteruno* {SeGA-IL: error}
(Do you think that it is hot?)
U3: *<Atsukuha naidesuyo>* (I do not think that it is thick.) [*Atsukuha naidesuyo*] (I do not think that it is hot.)
S3: *Douka Atsuku Ringono kotowo Oshieteyo* {ELIZA: semi correct} (Please teach me the apple thickly.)
U4: *<Ringoga Atsuinjya naindesuyo>* (The apple is not thick.) [*Ringoga Atsuinjya naindesuyo*]
(The apple is not hot.)
S4: *Atsuito Kangaeteruno* {ELIZA: correct}
(Do you think that the apple is thick?)
U5: *Kangaete naidesu* (I do not think so.)
S5: *Atsuikoto nandane* {ELIZA: correct}
(It is hot, isn't it?)
U6: *Chigaimasu* (No, it isn't.)
S6: *Meiwakude nakereba Okoranaidene* {SeGA-IL: correct}
(Please do not get angry if you don't mind.)
U7: *Daijyoubudesu* (I'm OK.)
S7: *ttute Henjyanai* {ELIZA: correct}
(Is it strange?)
U8: *Henjya naidesuyo* (It isn't strange.)
S8: *Ringoga Henjyatte kotonanokana* {ELIZA: semi-correct} (Then is the apple strange?)
U9: *Soujya naidesuyo* (No, it isn't.)
S9: *Tashikani Henjyayone* {ELIZA: semi-correct}
(It is certainly strange.}
U10: *Hendeha arimasenyo* (It isn't strange.)

< >: This sentence includes speech recognition errors.
[ ]: Correct utterance { }: Type of result
Italic characters are Romanized Japanese.
A capital word means an independent word in Japanese.

Fig. 3 Example of Dialogue in Japanese

At U6 and S6 of Fig. 3, a user says "No, it is not.", and then the system replies "Please do not get angry if you don't mind." This reply is natural. The system learned SeGA-IL-type rules from the user's previous utterances using SeGA-IL. It used them to generate this reply.

Figure 4 shows an English example. At U6 and S6 of Fig. 4, a user utters "No.", and then the system replies "Why 'no'?" This reply is very natural. The system learned SeGA-IL-type rules from the user's previous utterances using SeGA-IL in the same way as in the case of Japanese.

The system can automatically generate this reply using these rules.

U1: <There is not> [No it is not]
S1: Please go on.{SeGA-IL: semi-correct}
U2: What should I say
S2: I all things said that so far.{ELIZA: correct}
U3: But I don't understand you
S3: Do you like cinema {ELIZA: semi-correct}
U4: Yes I do
S4: Please tell me some more about this. {ELIZA: semi-correct}
U5: <I like to imagine a good movie in the cinema>
[I like to watch a good movie in the cinema]
S5: Does talking about this bother you? {ELIZA: semi correct}
U6: No
S6: Why 'no'? {SeGA-IL: correct}
U7: <Because it is you travel> [Because it is neutral]
S7: You're not really talking about me – are you?
{SeGA-IL: semi-correct}
U8: Now I'm not

< >: This sentence includes speech recognition errors.
[ ]: Correct utterance { }: Type of result

Fig. 4 Example of Dialogue in English

These results and considerations verify that this method has high generality for Japanese and English.

Moreover, in another experiment, we confirm that the correct rate using only GA-IL is 12.0% and that the correct rate using only SeGA-IL is 17.8%. Therefore, the correct rate of the system with SeGA-IL improved 5.8 points compared with that of the system with GA-IL. We consider that SeGA-IL is effective for this application.

5. Conclusion

In this paper, we have proposed SeGA-ILSD for a spoken dialogue system for chat. We described outlines and details of this method. Then we carried out experiments to evaluate the generality of this method using Japanese and English. The experimental system based on this proposed method shows the same performance for both languages. Results of this experiment show that this proposed method has high generality for Japanese and English.

Future works will include experiments using other languages such as Spanish, Chinese, Polish, Italian, Swedish, and so on to evaluate the generality of this method for other languages.

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